4 Power Question 2017: The Paral-Elle Universe

Instructions: The power question is worth 50 points; each part's point value is given in brackets next to the part. To receive full credit, the presentation must be legible, orderly, clear, and concise. If a problem says "list" or "compute," you need not justify your answer. If a problem says "determine," "find," or "show," then you must show your work or explain your reasoning to receive full credit, although such explanations do not have to be lengthy. If a problem says "justify" or "prove," then you must prove your answer rigorously. Even if not proved, earlier numbered items may be used in solutions to later numbered items, but not vice versa. Pages submitted for credit should be NUMBERED IN CONSECUTIVE ORDER AT THE TOP OF EACH PAGE in what your team considers to be proper sequential order. PLEASE WRITE ON ONLY ONE SIDE OF THE ANSWER PAPERS. Put the TEAM NUMBER (not the team name) on the cover sheet used as the first page of the papers submitted. Do not identify the team in any other way.

Elle is a student at Springfield High School. One day, she's escorted from her math class by a friendly space alien, who takes her to a parallel universe to learn math the way they do math, which is different than the way we do math. In the parallel universe, students are taught two operations: $\oplus$ and $\odot$. Elle notices that

$$a \oplus b = \min(a, b) \quad \text{and} \quad a \odot b = a + b$$

for all real numbers a and b. Elle decides to investigate the properties of these two new operations.

Throughout this Power Question, $a \oplus b$ and $a \odot b$ will always denote parallel-universe addition and multiplication, respectively, while $a + b$ and $a \cdot b$ will denote addition and multiplication in the usual sense. The standard order of operations ($\odot$ first, then $\oplus$) will apply, although parentheses are sometimes included for clarification. All variables refer to real numbers unless otherwise specified.

1. **a.** Determine whether there is a real number $\mathcal{O}$ such that $\mathcal{O} \oplus x = x$ for all x. [2 pts]

b. Show that there is a real number $\mathcal{I}$ such that $\mathcal{I} \odot x = x$ for all x. [2 pts]

2. Show that $\odot$ distributes over $\oplus$; that is, $a \odot (b \oplus c) = (a \odot b) \oplus (a \odot c)$ for all a, b, c. [3 pts]

3. Find all solutions to each of the following equations:

a. $(x \odot x) \oplus (2 \odot x) \oplus 1 = x \oplus 2$; [3 pts]

b. $(x \odot x) \oplus (-1) = (x \oplus 1) \odot (x \oplus (-1))$. [3 pts]

Next, Elle investigates the properties of exponents and polynomials. She uses the notation

$$a^{\odot b} = \underbrace{a \odot a \odot a \odot \cdots \odot a}_{b \text{ copies of } a},$$

where a is a real number and b is a positive integer. Furthermore, by definition, $a^{\odot 0} = 0$. Elle wonders if exponentiation works in the parallel universe the way it works in ours.

4. Show that for any real numbers x and y, and any positive integer n:

a. $(x \odot y)^{\odot n} = x^{\odot n} \odot y^{\odot n}$; [2 pts]

b. $(x \oplus y)^{\odot n} = x^{\odot n} \oplus y^{\odot n}$; [2 pts]

c. $(x \oplus y)^{\odot 2} = x^{\odot 2} \oplus y^{\odot 2} = (x^{\odot 2}) \oplus (x \odot y) \oplus (y^{\odot 2})$. [3 pts]

5. **a.** Show that for any a, there is exactly one solution to the equation $x^{\odot 2} = a$. This is called the **parallel square root of a** and is denoted by $\sqrt[\circ]{a}$. [2 pts]

b. Show that for any a, there is exactly one solution to the equation $a \odot x = \mathcal{I}$, where $\mathcal{I}$ is defined in Problem 1b. This is called the **parallel reciprocal of a** and is denoted by $\widehat{a}$. [2 pts]

c. Show that for any a, $\sqrt[\circ]{\widehat{a}} = \widehat{\sqrt[\circ]{a}}$. [3 pts]

In the parallel universe, a **parallel-universe polynomial (PUP)** of degree n is a finite "sum" (i.e., parallel-universe addition) of $n+1$ parallel products whose exponents decrease from n to 0, inclusive:

$$f(x) = (a_n \odot x^{\odot n}) \oplus (a_{n-1} \odot x^{\odot(n-1)}) \oplus \cdots \oplus (a_1 \odot x^{\odot 1}) \oplus (a_0 \odot x^{\odot 0}),$$

where each a_i is a real number. As a convenient shorthand, $x^{\odot 1}$ will simply be written as x and $a_0 \odot x^{\odot 0}$ will simply be written as a_0. For example:

- $f(x) = (5 \odot x) \oplus 3$ is a degree-1 PUP, but not a degree-2 PUP.
- $f(x) = (1 \odot x^{\odot 2}) \oplus 1$ is not a PUP at all, because the x term is missing.
- $f(x) = (0 \odot x^{\odot 2}) \oplus (0 \odot x) \oplus 0$ is a degree-2 PUP.
- $f(x) = (x^{\odot 2}) \oplus x \oplus 1$ is not a PUP; it should instead be written as $(0 \odot x^{\odot 2}) \oplus (0 \odot x) \oplus 1$.

Elle now examines the properties of degree-2 (quadratic) PUPs. In the parallel universe, a **root** of a PUP f is a real number x such that $f(x) = 0$.

6. In our universe, a quadratic equation has either zero, one, or two distinct real roots.

- **a.** Find a quadratic PUP that has exactly one root. [2 pts]
- **b.** Could a quadratic PUP have zero roots? Justify your answer. [2 pts]
- **c.** Could a quadratic PUP have infinitely many distinct roots? Justify your answer. [2 pts]
- **d.** Suppose that $f(x)$ is a PUP of positive degree. Prove that if f has more than one root, then it has infinitely many distinct roots. [3 pts]

7. Suppose $f(x) = (a \odot x^{\odot 2}) \oplus (b \odot x) \oplus c$ is a quadratic PUP that has exactly one root. Find a formula for this root in terms of a, b, and c. Express the formula using only the operations $\oplus$, $\odot$, parallel-universe exponentiation, parallel square roots, and parallel reciprocals. [5 pts]

Suppose $f(x) = (a \odot x^{\odot 2}) \oplus (b \odot x) \oplus c$ is a quadratic PUP. A **factorization** of f is an ordered triple of real numbers (k, r, s) such that $r \leq s$, and

$$f(x) = k \odot (x \oplus r) \odot (x \oplus s),$$

for all real numbers x.

8. Prove that if (k, r, s) is a factorization of f, then:

- **a.** $k = a$, and [2 pts]
- **b.** $r \odot s = c \odot \widehat{a}$. [2 pts]

9. Prove that every quadratic PUP has exactly one factorization. [5 pts]